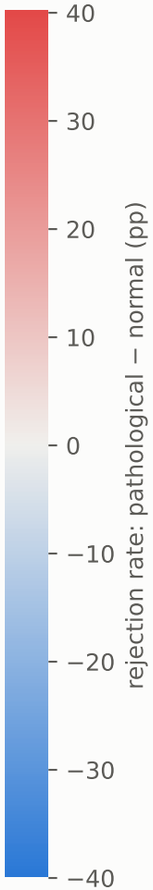
E2s Equal-noise gap at 10 dB, noise scaled to the S1+S2 power (matched threshold, mean of 3 seeds)

	CirCor		ZCHSound				BMD-HS†					PASCAL†		Synthetic rhythm	
	Systolic murmur	Diastolic murmur*	ASD	VSD	PFO	PDA	AS	AR	MR	MS	Multi-valve	Murmur	Extrasystole	Premature (synth.)	AF-like (synth.)
B1 Springer SQI	**-9** CI≠0	+12	**-9** CI≠0	+3	-1	+4	-1	-14	+15	-5	+5	+8	+8	**+22** CI≠0	**+24** CI≠0
B2 Das RF	-1	-1	-0	-0	-4	+2	+0	-5	+0	+5	+4	+8	+3	**+5** CI≠0	**+6** CI≠0
B3 CardioPHON	**-7** CI≠0	-4	**+4** CI≠0	+2	**+6** CI≠0	+4	**+6** CI≠0	-4	+2	+2	+3	+2	+3	**+8** CI≠0	**+9** CI≠0
B4 CNN	-2	+6	**+6** CI≠0	**+3** CI≠0	**+8** CI≠0	**+7** CI≠0	-1	-3	+0	+0	-0	+5	+2	+2	+2
B5 CRNN	**+5** CI≠0	+3	**+4** CI≠0	**+3** CI≠0	**+8** CI≠0	**+5** CI≠0	+0	-1	+1	+0	+0	+6	-1	+1	+2
B6 PCGKolect	**+5** CI≠0	**+30** CI≠0	+0	-5	**+10** CI≠0	+9	+2	-2	+2	**+2** CI≠0	**+2** CI≠0	-5	+1	**+51** CI≠0	**+30** CI≠0
B7 SNR rule	**-4** CI≠0	**+24** CI≠0	-3	**+7** CI≠0	+4	**+17** CI≠0	+8	+3	+8	+6	**+12** CI≠0	+2	+0	**-9** CI≠0	**-9** CI≠0
B8 PCGKolect-v3	**+8** CI≠0	+20	+0	+0	-1	+1	+0	-0	+0	-0	-0	-0	-3	**+6** CI≠0	+1
H-AND Das∧B4	+1	+3	+3	+1	+2	+4	+1	-6	-1	+6	+4	**+6** CI≠0	+4	**+6** CI≠0	**+7** CI≠0
H-AND Das∧B5	+3	+4	+3	+1	+2	+3	+1	-7	-2	+5	+3	**+6** CI≠0	+2	**+5** CI≠0	**+7** CI≠0
H-CASCADE Das→B4	-2	+7	**+4** CI≠0	+3	**+4** CI≠0	+6	+1	-3	-1	+2	+2	+5	+4	+2	**+3** CI≠0
H-CASCADE Das→B5	+3	+5	+2	+3	+3	+5	+1	-3	-0	+2	+2	+7	+5	+1	+2
H-FUSE Das+B4	-3	+2	**+5** CI≠0	+2	**+8** CI≠0	+5	-2	-4	+1	+1	-0	**+6** CI≠0	+3	**+3** CI≠0	**+4** CI≠0
H-FUSE Das+B5	+0	+10	**+4** CI≠0	**+3** CI≠0	**+7** CI≠0	**+6** CI≠0	-0	-5	**+3** CI≠0	+1	-0	+7	+1	+0	+1
HD1 avg Das,B4	+0	+1	+3	+2	**+4** CI≠0	+4	-1	-5	-1	+1	-0	**+6** CI≠0	+3	**+6** CI≠0	**+7** CI≠0
HD2 min Das,B4	+0	+0	+3	+2	+3	+3	-1	-4	+0	+1	-0	**+6** CI≠0	+2	**+6** CI≠0	**+7** CI≠0
HD3 LR Das+B4 emb.	-0	+16	+1	+3	**+7** CI≠0	+3	-3	-8	+1	+0	-0	**+7** CI≠0	-0	**+12** CI≠0	**+13** CI≠0
HD4 λ0.1 (chosen)	**-4** CI≠0	-3	**+4** CI≠0	+2	**+7** CI≠0	+3	-1	-3	+0	+2	+1	+5	+0	**+8** CI≠0	**+9** CI≠0
HD4 λ0.3	-3	-0	+2	+2	**+3** CI≠0	+2	-0	-1	+0	+0	-0	**+5** CI≠0	+2	**+9** CI≠0	**+10** CI≠0
HD4 λ1.0	-4	+2	**+3** CI≠0	**+4** CI≠0	**+6** CI≠0	+3	+0	+0	+1	+1	+1	**+5** CI≠0	+3	**+11** CI≠0	**+12** CI≠0



Same beats and noise clips as E2; the SNR reference is the power of the clean window inside its S1 and S2 segments, so murmur energy does not raise the noise level. Bold/"CI≠0" = CI excludes 0 in ≥ 2 of 3 seeds.